

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801C - HIGHER SURVEYING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Higher surveying primarily differs from conventional surveying due to its emphasis on:
 - A) Chain and compass methods
 - B) High-precision measurements and advanced instrumentation
 - C) Plane table techniques
 - D) Small-area surveys
 - 2 One fundamental requirement of higher surveying is:
 - A) Redundancy of observations for accuracy
 - B) Approximate observations
 - C) Use of only graphical methods
 - D) Elimination of computations
 - 3 A geocentric coordinate system has its origin at the:
 - A) Local benchmark
 - B) Survey station
 - C) Centre of the Earth
 - D) Mean sea level
 - 4 The principle of least squares aims to:
 - A) Minimize the number of observations
 - B) Minimize the sum of squares of residuals
 - C) Maximize observation errors
 - D) Eliminate gross errors automatically
 - 5 Photogrammetry involves obtaining measurements from:
 - A) Satellite signals
 - B) Ground observations only
 - C) Photographs
 - D) Radar waves

- 6 The scale of a vertical aerial photograph mainly depends on:
 - A) Camera focal length and flying height
 - B) Ground slope only
 - C) Weather conditions
 - D) Film type
- 7 Relief displacement in aerial photographs is caused by:
 - A) Camera tilt
 - B) Earth's curvature
 - C) Height of objects above datum
 - D) Atmospheric refraction
- 8 LiDAR determines distance primarily by measuring:
 - A) Signal frequency change
 - B) Travel time of laser pulse
 - C) Signal strength
 - D) Phase angle only
- 9 Airborne LiDAR systems are commonly used for:
 - A) Underground surveys
 - B) Large-area topographic mapping
 - C) Indoor measurements
 - D) Small construction layouts
- 10 Sounding surveys are carried out to determine:
 - A) Ground elevations
 - B) Water depth below water surface
 - C) Soil strength
 - D) Wave height
- 11 Echo sounding works on the principle of:
 - A) Reflection of light
 - B) Refraction of sound waves
 - C) Reflection of sound waves
 - D) Absorption of energy
- 12 Higher surveying plays a critical role in smart city projects by:
 - A) Replacing civil engineers
 - B) Providing accurate spatial data for planning and management
 - C) Reducing population density
 - D) Eliminating GIS

Q2. Solve the following.

- A) What are the fundamental requirements of higher surveying? 6
- B) Explain platform requirements for higher surveying. 6

Q3. Solve the following.

- A) Write the different steps in positioning and direction survey. 6
- B) Explain in detail about close range and terrestrial photogrammetry mapping. 6

Q4. Solve Any Two of the following.

- A) Enlist different physical models of photogrammetry. 6
- B) Enlist different mathematical models of photogrammetry. 6
- C) Explain detailed account of LiDAR Technique and its applications in higher surveying. 6

Q5. Solve Any Two of the following.

- A) How the LiDAR data is processed to generate digital terrain models (DTM)? 6
- B) Write short notes on Sounding survey and Bathymetry survey? 6
- C) What are the applications of Higher Surveying techniques? 6

Q6. Solve Any Two of the following.

- A) What are the applications of RADAR from a geoscience perspective? 6
- B) State the different applications of Higher Surveying techniques for measurements of surface deformation and plate tectonic movement? 6
- C) State the different applications of Higher Surveying techniques for measurements of surface deformation and plate tectonic movement? 6

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